

Zerin Shaima Meem

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SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL

Machine Learning & AI: TensorFlow, Keras, scikit-learn, Hugging Face, NumPy, pandas, Matplotlib, Seaborn

Web & Backend: React, Angular, Node.js, MongoDB, Supabase, PostgreSQL

Tools & Platforms: Git/GitHub, AWS (RDS, Amplify), Google Colab, Postman, Jest, Playwright, Storybook

EDUCATION

Masters in Computer Science | AI-Concentration | Expected May 2027

CGPA: 3.89/4.00

Relevant Coursework: AI, Adv. AI, Deep Learning, Cloud Computing, Algorithms, Software Engineering

WORK EXPERIENCE

- Graduate Research Assistant at University of Nebraska Omaha** August 2025- Present
Advisor: Dr. Xiaoqian (Tiffany) Zhang, Assistant Professor of Computer Science
 - Investigated and evaluated rate adaptation algorithms (HPCC, DCQCN, DCTCP, TIMELY) and NS-3 simulated traffic scenarios, analyzing throughput and latency to identify configurations that improved network performance.
Funded by the Nebraska Research Initiative Award 2025-2026.
 - Designed and developed AI-driven congestion control experiments for large-scale storage systems, building model-driven decision pipelines that reduced congestion events response latency.
Funded by NSF CRII award 2025-2027.
- Software Engineer I at Samsung R&D Institute Bangladesh** January 2025- July 2025
 - Collaborated on a web-based semantic knowledge graph modeling tool that enables ontology creation, visualization, and data mapping. Built a persistent **real-time graph canvas** with **drag-and-drop** support for constructing and editing RDF-based knowledge structures; authored unit and end-to-end tests (Jest, Playwright) that increased automated test coverage by 20%, and delivered clean, isolated UI components through Git-based workflows.

PROJECTS

- Unsupervised Anomaly Detection on Network Traffic using Autoencoders** Python, TensorFlow/Keras, scikit-learn, pandas, NumPy, Matplotlib, Seaborn
 - Labeled attack data is scarce and constantly goes stale as new attack types emerge. Trained a TensorFlow/Keras autoencoder exclusively on normal traffic from the NSL-KDD dataset, flagging anomalies by reconstruction error and achieving **0.956 ROC-AUC** and **86%** accuracy without ever seeing attack examples during training.
- Comparative Analysis of Deep Learning Models for IMDB Sentiment Classification** Python, TensorFlow, Keras, scikit-learn, Hugging Face Transformer, pandas
 - Benchmarked Logistic Regression (TF-IDF) against LSTM, Text CNN, and DistilBERT on IMDB movie reviews; Logistic Regression achieved the strongest **0.871 F1-score** while training **5-10x faster** than the neural models, showing that model complexity doesn't guarantee better performance.
- Portfolio Builder** React, Node.js, AWS RDS/PostgreSQL, AWS Amplify, JavaScript
 - Built a full-stack web app for creating accounts and publishing shareable portfolio pages; integrated AWS RDS /PostgreSQL for relational data storage and supported authentication workflows, responsive UI, and backend APIs.
- CareTrack-Connect** JavaScript, Node.js, Supabase, SQL, Vite
 - Built a digital health tracking app enabling patients to submit wellness questionnaires and doctors to review patient data; developed backend APIs and database schema for structured patient/questionnaire data.

PUBLICATION

- A CSI-based Position Independent Gesture Recognition System Using DL** IEEE WIECON-ECE 2024
 - An LSTM-based classifier achieving **81.1%** accuracy for position independent gesture recognition using WiFi CSI data. Designed data pipeline: ESP-32 CSI data collection → preprocessing → ML/DL model evaluation.
- Spot Light: Disaster Safety Assistance Mobile Application** IEEE ICCPCT 2024
 - An application enabling emergency messaging, fund transfers, and rescue center mapping to support disaster-affected communities.

ACHIEVEMENTS

- GRACA Scholarship (\$5,000)**, UNO, Summer 2026, for project titled "Adaptive Rate Control for UAV Communication Networks Using Deep Reinforcement Learning."
- Honorable Mention**, UNO C-Bytes Research Showcase Competition.
- Solved 3000+ programming problems | [Codeforces- 1607\(Expert\)](#) | [CodeChef- 1788\(3*\)](#) | 3-time ICPC Dhaka Regional participant | **Globally ranked 9th** in ICPC AlgoQueen 2024